Power

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare how quickly energy is transferred.

SUCCESS CRITERIA - I can...

1. Distinguish energy in J from power in W.
2. Use power=energy/time.
3. Explain equal work done at different rates.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Supplied data: m=0.3 kg, drop h=0.5 m, final speed 1.5 m/s from rest, g=9.8. Find mgh, translational K and the fraction appearing as translation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: P_{avg} = W/Δt = energy transferred/Δt; P = Fv. Second equation: force parallel to velocity. Power units: 1 W = 1 J/s.

Worked reasoning: Two motors each transfer 120 J. One takes 3 s, the other 6 s. Their average powers are 40 and 20 W; total transferred energy is equal.

Your turn: Draw equal energy-transfer arrows and label the different times and powers.

Case	Energy (J)	Time (s)	Power (W)
1	____	____	____
2	____	____	____

Marzano check - circle: ① ② ③

03 Read and interpret the model

Two motors each transfer 120 J. One takes 2 s; the other 6 s. Which has greater average power?

Motor	Energy	Time
A	120 J	2 s
B	120 J	6 s

Marzano check - circle: ① ② ③

04 Apply the physics

A 60 kg climber gains 4 m of vertical height in 8 s. Use g = 9.8 N/kg. Calculate gravitational energy gain and average mechanical power. Label the energy and time in the table.

Case	Energy (J)	Time (s)	Power (W)
1	____	____	____
2	____	____	____

Marzano check - circle: ① ② ③

05 Explain your reasoning

A 100 W label is described as “100 joules.” Correct the statement and give time information that would make an energy calculation possible.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A 4 kg object slows from 8 to 6 m/s. Find net work and check its sign.
Show the energy changes and transfers; label units.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

07 Exit check

A constant 200 N forward force acts at steady 3 m/s. Find mechanical power and work in 5 s.

Case	Energy (J)	Time (s)	Power (W)
1	____	____	____
2	____	____	____

Marzano check - circle: ① ② ③

08 STRETCH: test the model

A motor lifts a 20 kg load 3 m in 4 s at constant speed; g=10. Find useful power. If efficiency is 75%, find input power and energy not delivered to the load.

Case	Energy (J)	Time (s)	Power (W)
1	____	____	____
2	____	____	____

Marzano check - circle: ① ② ③